SNEKBOROS: VISION WORLD MODELS FOR COMPETITIVE SNAKE PLAYING

TEAM 5

1. Methodology

1.1. **Data.** Let $\mathrm{L} \in \mathbb{N}_{\geq 1}$ be a grid extent, and $\mathrm{U} \in \mathbb{N}_{\geq 1}$ be an upsampling factor, and $\mathrm{S} \in \mathbb{N}_{\geq 1}$ be a number of snakes.

Definition 1 (Grid)**.**

$$\mathcal{G} = [1..\mathrm{L}]^2$$

Definition 2 (Action Space)**.**

$$\mathcal{A} = \{(0,0), (0,1), (0,-1), (-1,0), (1,0)\}$$

Definition 3 (Movement Space)**.**

$$\mathcal{M} = \mathcal{A} \setminus \{(0,0)\}$$

Definition 4 (Snake Space)**.**

$$\mathcal{S} = \bigcup_{n \geq 2} \left\{ \boldsymbol{s} \in \left(\mathbb{Z}^2\right)^n \;\middle|\; \forall i \in [1..n-1](s_{i+1} - s_i \in \mathcal{M}) \right\}$$

Definition 5 (Food Space)**.**

$$\mathcal{F} = \mathcal{G} \cup \{\perp\}$$

Definition 6 (Action Oracle)**.**

$$\delta : \mathcal{S} \times \mathbb{N} \to \mathcal{A}$$

Definition 7 (Snake Occupancy)**.** Let $\boldsymbol{S} \in \mathcal{S}^*$.

$$\mathcal{O}(\boldsymbol{S}) = \{s_i \mid \boldsymbol{s} \in \boldsymbol{S} \wedge i \in [1..|\boldsymbol{s}|]\}$$

Definition 8 (Pairing Function)**.**

$$\rho : \mathcal{G}^{2\mathrm{S}} \to (\mathcal{G} \times \mathcal{G})^{\mathrm{S}}$$
$$\rho(\boldsymbol{x}) = ((x_{2i-1}, x_{2i}))_{i=1}^{\mathrm{S}}$$

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Algorithm 1 GAME

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S ~ ρ ∘ U({s ∈ G^{2S} | ∀i ∈ [1..S](s_{2i} − s_{2i−1} ∈ M) ∧ |{s_i}_{i=1}^{2S}| = 2S})
E ← G \ O(S)
if E = ∅ then
    f ←⊥
else
    f ← U(E)
end if
t ← 0
while |S| > 0 do
    i_f ← 0
    for all s ∈ S do                                   ▷ Translate snakes
        d ← δ(s, t)
        if d = (0, 0) then
            h ← 2s_1 − s_2
        else
            h ← s_1 + d
        end if
        s ← h ‖ (s_i)_{i=1}^{|s|−1}
    end for
    for all s ∈ S do                                   ▷ Grow snakes
        if s_1 = f then
            s ← s ‖ (2s_{|s|} − s_{|s|−1})
            i_f ← 1
        end if
    end for
    D ← ∅
    B ← {s_i | s ∈ S ∧ i ∈ [2..|s|]}
    for all s ∈ S do
        if s_1 ∉ G ∨ s_1 ∈ B then
            D ← D ∪ {s}
        end if
    end for
    H ← {g ∈ G | ∃c ⊆ S(|c| ≥ 2 ∧ ∀s ∈ c(s_1 = g))}
    for all h ∈ H do
        C ← {s ∈ S | s_1 = h}
        w ← U(arg max_{s∈C} |s|)
        D ← D ∪ (C \ {w})
    end for
    for all d ∈ D do                                   ▷ Kill snakes
        S ← S \ {d}
    end for
    if i_f = 1 then                                    ▷ Respawn food
        E ← G \ O(S)
        if E = ∅ then
            f ←⊥
        else
            f ← U(E)
        end if
    end if
    t ← t + 1
end while
```

1.2. **Architecture.** Let $\mathrm{R} \in \mathbb{N}_{\geq 1}$ be a representation space dimensionality, and $\mathrm{Z} \in \mathbb{N}_{\geq 1}$ be a latent space dimensionality, and $\mathcal{X} = \mathbb{N}_{\leq 255}^{\mathrm{UL} \times \mathrm{UL} \times 3}$ be a raw space, and $\mathcal{R} = \mathbb{R}^{\mathrm{R}}$ be a representation space, and $\mathcal{Z} = \mathbb{R}^{\mathrm{Z}}$ be a latent space, and $\mathbb{P}_{\mathcal{Z}}$ be a probability measure on $\mathcal{Z}$, and $\gamma \in \mathbb{R}_{[0,1]}$ be a discount factor, and $T \in \mathbb{N}_{\geq 1}$ be a planning horizon.

Definition 9 (Colouring Function)**.**

$$\mathcal{C} : \mathcal{G} \times \mathcal{S}^* \times \mathcal{F} \to \mathbb{N}_{\leq 255}^3$$

$$\mathcal{C}((i,j), \boldsymbol{S}, f) = \begin{cases} (128, 128, 255), & \text{if } |\boldsymbol{S}| \geq 1 \wedge (i,j) = s_{1,1} \\ (0, 0, 255), & \text{if } |\boldsymbol{S}| \geq 1 \wedge \exists k \in [2..|\boldsymbol{s}_1|]((i,j) = s_{1,k}) \\ (255, 128, 128), & \text{if } \exists r \in [2..|\boldsymbol{S}|]((i,j) = s_{r,1}) \\ (255, 0, 0), & \text{if } \exists r \in [2..|\boldsymbol{S}|], \exists k \in [2..|\boldsymbol{s}_r|]((i,j) = s_{r,k}) \\ (0, 255, 0), & \text{if } f \neq \perp \wedge (i,j) = f \\ (0, 0, 0), & \text{otherwise} \end{cases}$$

Definition 10 (Rasterisation Function)**.**

$$\mathcal{D} : \mathcal{S}^* \times \mathcal{F} \to \mathcal{X}$$

$$\mathcal{D}(\boldsymbol{S}, f) = \left(\mathcal{C}\left(\left(\left\lceil \frac{i}{\mathrm{U}} \right\rceil, \left\lceil \frac{j}{\mathrm{U}} \right\rceil \right), \boldsymbol{S}, f \right) \right)_{(i,j) \in [1..\mathrm{UL}]^2}$$

Definition 11 (Encoder)**.**

$$\phi : \mathcal{X} \to \mathcal{R}$$

Definition 12 (World Model)**.**

$$\mathscr{F} : \mathcal{R} \times \mathcal{A} \times \mathcal{Z} \to \mathcal{R}$$

Definition 13 (Energy)**.**

$$\mathscr{C} : \mathcal{R} \to \mathbb{R}_{\geq 0}$$

Definition 14 (Discounted Sum Planner)**.**

$$f : \mathcal{X} \to \bigsqcup_{t \geq 1} \mathcal{A}^t$$

$$f(x) = \mathcal{U}\left(\arg\min_{\boldsymbol{a} \in \mathcal{A}^T} \mathbb{E}_{\boldsymbol{z} \sim \mathbb{P}_{\mathcal{Z}}^{\otimes T}} \left[\sum_{t=1}^{T} \gamma^{t-1} \mathscr{C}\left(\left(\bigcirc_{i=1}^{t} \mathscr{F}(\cdot, a_i, z_i) \right)(\phi(x)) \right) \right] \right)$$

2. Appendix

Notation 1. Let x be a set, and $\boldsymbol{y}$ be a sequence.

$$x \in \boldsymbol{y} \equiv x \in \mathrm{im}(\boldsymbol{y})$$

Notation 2. Let $\boldsymbol{x}, \boldsymbol{y}$ be sequences.

$$\boldsymbol{x} \subseteq \boldsymbol{y} \equiv \mathrm{im}(\boldsymbol{x}) \subseteq \mathrm{im}(\boldsymbol{y})$$

Notation 3. Let $\boldsymbol{x}$ be a sequence, and y be a set.

$$\boldsymbol{x} \setminus y = (x_i)_{i \in \{j \in \mathrm{dom}(\boldsymbol{x}) | x_j \notin y\}}$$

Notation 4. Let $\boldsymbol{x}$ be a sequence.

$$|\boldsymbol{x}| = |\mathrm{dom}(\boldsymbol{x})|$$

Notation 5. Let $n \geq 1$, and X be a set, and $f_1, \ldots, f_n : X \to X$.

$$\bigcirc_{i=1}^{n} f_i = f_n \circ \cdots \circ f_1$$